Table 1 lists some of the important options of the *ping* command.

Table 1

Arguments	Description
-h	This argument will help you to list options related to ping.
-b	Allows pinging a broadcast address.
-q	Quiet output. Nothing is displayed except the summary lines at startup time and when finished.
-t	Sets the IP time to Live.
-v	Shows details in verbose mode.
-V	Will only show version and exits.
-c	This indicates the number of echo requests to send out.
-a	Audible ping.
-i	Waits for specified seconds before sending a new packet. The default value is one second.

For more options with detailed explanations, refer to the manual page of the *ping* command.

Scanning network ports

Nmap is an open source port scanner tool that administrators use to rapidly scan large networks; it can also do more intensive port scans on individual hosts.

What does Nmap do?

Nmap uses raw IP packets in novel ways to determine what hosts are available on the network, what services (application name and version) those hosts are offering, what operating systems (and OS versions) they are running, what type of packet filters/firewalls are in use, and dozens of other characteristics.

The Nmap executable is provided by the Nmap package. You can install the Nmap package by executing the following command:

```
# yum -y install nmap or # rpm -ivh nmap
```

Figure 3 shows Nmap scanning 192.168.123.2 (a local virtual machine). The *-n* option tells Nmap to display host information numerically, not using DNS. As Nmap discovers each host, it scans privileged TCP ports looking for services. It displays the MAC address, with the corresponding network adapter manufacturer, of each host. Nmap is generally used to scan the complete network.

```
[root@ipa ~]# nmap -n 192.168.123.2

Starting Nmap 6.40 ( http://nmap.org ) at 2019-02-23 00:18 IST
Nmap scan report for 192.168.123.2
Host is up (0.000047s latency).
Not shown: 990 closed ports
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
443/tcp   open  https
3128/tcp open  squid-http
5000/tcp open  upnp
8008/tcp open  http
8009/tcp open  ajp13
8080/tcp open  http-proxy
8443/tcp open  https-alt
9090/tcp open  zeus-admin
MAC Address: 52:54:00:11:F4:50 (QEMU Virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.13 seconds
[root@ipa ~]#
```

Figure 3: Nmap is scanning a local virtual machine

```
[root@ipa ~]# nmap  -n -sn 192.168.123.2

Starting Nmap 6.40 ( http://nmap.org ) at 2019-02-23 00:24 IST
Nmap scan report for 192.168.123.2
Host is up (0.00039s latency).
MAC Address: 52:54:00:11:F4:50 (QEMU Virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.04 seconds
[root@ipa ~]#
```

Figure 4: Nmap is not scanning the ports here

A port scan can be disabled by using the *-sN* option. Administrators use this option when they want to quickly see which hosts are running on a network. You can refer to Figure 4, in which Nmap is scanning the virtual machine without scanning the port. This option is very useful when you just want to check which machines are up in your network.

There is one more interesting option, the *-sU* option, that tells Nmap to perform a UDP port scan. This scan takes substantially more time to run than the default TCP port scan. It is useful when an administrator wants a more complete picture of what services a host exposes to the network.

There are many other Nmap options that you can go through from the manual page.

Communicating with a remote service

Ncat is a troubleshooting tool that allows administrators to communicate directly with a port. Ncat can use either TCP or UDP to communicate with a network service, and it also supports SSL communication. Ncat is invoked as either the *nc* or *ncat* commands. To use this utility, download the *nmap-ncat* package on your Linux systems.

Ncat acts as a server when it is invoked with the *-l* option. This is known as 'listen mode' for Ncat. The *-k* option allows Ncat to keep the port open to listen for more connections when used in this mode.

The default behaviour of Ncat in listen mode is to display any text that it receives over the network to the screen. The *-e* argument causes Ncat to pass incoming network traffic to the command specified with this option, as shown below. The following command will launch Ncat to listen on port 4343. It will pass the traffic on the bash shell. On my test remote system, the following command has been executed:

```
# nc -l 4343 -e /bin/bash
```